Neel Gali

Software Engineer (Java) in Test

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | AI-Augmented Multi-Platform Quality Engineering

- Architected and implemented an **AI-augmented test automation framework** using **Java** for **CTV** and mobile platforms, accelerating test authoring velocity by 35%.
- Developed advanced **prompt engineering** strategies with **LLM-based assistants** like **GitHub Copilot** to generate complex test cases, reducing manual effort by 40%.
- Integrated automated **QoE metrics** validation for **video playback** and **streaming resiliency** using **ffmpeg** across **Roku** and **Fire TV**, improving anomaly detection by 25%.
- Designed and delivered **device-level testing** for **DRM flows** and **SSAI/CSAI ad insertion** on **Samsung Tizen** and **LG webOS** using **Appium**, ensuring 99.9% content protection.
- Optimized **CI/CD pipelines** to incorporate **shift-left testing** with **Java** and **TestNG**, achieving continuous quality for enterprise platforms and reducing defect escape rate by 20%.
- Championed evaluation and integration of **Cursor** for **root-cause analysis** and **log/telemetry summarization** within existing frameworks, decreasing defect triage time by 30%.
- Developed **reusable libraries** and utilities for **captions/subtitles** and **content protection** validation across **iOS/Android mobile apps** and web, standardizing testing efficiency by 25%.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Cloud-Native Test Automation Platform

- Developed and maintained a robust **Java**-based **test automation framework** for enterprise **API** and **UI testing** on **AWS**, improving test coverage by 40%.
- Implemented comprehensive **integration testing** for microservices using **REST-Assured** and **JUnit** within a **Jenkins CI/CD pipeline**, reducing post-deployment issues by 15%.
- Engineered **performance testing** suites using **JMeter** and **Locust** to simulate high user loads on **cloud-native applications**, identifying critical bottlenecks and improving system reliability by 20%.
- Created and extended **reusable libraries** for data management and assertions in **Selenium** and **Appium** test suites, significantly increasing test stability and reducing flakiness by 30%.
- Integrated **GitHub Copilot** into the development workflow for accelerating **Java** test authoring and code reviews, boosting team productivity by 18%.
- Designed and executed **regression testing** strategies for complex financial applications, ensuring data integrity and system consistency, which prevented 10+ critical production defects annually.
- Managed test data and environments using **Docker** and **PostgreSQL** for consistent, isolated testing, reducing environment setup time by 50% for various projects.
- Automated reporting and dashboard generation for test execution results, providing real-time insights into quality status, reducing manual reporting effort by 70%.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Enterprise Application Quality Assurance

- Contributed to a core **Java** application development team, implementing new features and maintaining existing modules, improving system functionality and user experience by 15%.
- Developed and executed manual and automated test cases for web applications using **Selenium WebDriver**, identifying critical bugs and ensuring product quality for 5+ releases.
- Designed and implemented robust **RESTful APIs** for data exchange, adhering to security standards and improving application interoperability by 20%.
- Collaborated with QA engineers to improve test coverage for various modules, writing test scripts and performing data validation against **PostgreSQL** databases, enhancing data accuracy by 98%.
- Streamlined debugging and issue resolution processes by analyzing application logs and performance metrics, reducing mean time to repair (MTTR) by 25%.
- Participated in **Agile** sprint planning and code reviews, ensuring adherence to coding standards and best practices, which improved code quality by 10% across projects.

Skills

Languages: python, Java, Typescript, Javascript
Frameworks & Tools: Copilot, Cursor, Selenium, Appium, TestNG, JUnit, JMeter, Locust, Jenkins, ffmpeg, Docker
Cloud & Infra: AWS, PostgreSQL
Concepts: API Testing, UI Testing, Mobile Testing, CTV/SmartTV Testing, Integration Testing, Performance Testing, Reliability Testing, Device-level Testing, CI/CD, Microservices, RESTful APIs, Agile, Prompt Engineering, Root-Cause Analysis, Parallelization, Regression Testing, QoE Metrics, Video Playback, DRM Flows, SSAI/CSAI Ad Insertion

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India